

Code Layout, Readability and Reusability

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Team ID	SWTID-2026-9792
Project Name	FitBuddy – AI Fitness Plan Generator

Code Layout, Readability and Reusability

FitBuddy uses a modular structure so major responsibilities are separated instead of being placed in one large file.

Path	Purpose
app/main.py	Creates the FastAPI application, middleware, static files, templates and router.
app/routes.py	Defines user, workout, feedback and admin routes.
app/database.py	Defines SQLAlchemy models and database helper functions.
app/gemini_generator.py	Generates personalized 7-day workout plans.
app/gemini_flash_generator.py	Generates concise nutrition/recovery tips.
app/updated_plan.py	Generates revised plans from original plans and feedback.
app/admin_auth.py	Verifies admin credentials loaded from environment variables.
templates/*.html	Contains the user and admin interface pages.
static/images/*	Contains visual assets used by the interface.

Readability Practices

The code uses descriptive function names, modular files, comments separating application sections, environment-based secrets, reusable database helper functions and server-rendered templates.